

FinGear AI: An Explainable Financial Digital Twin for Financial Health Scoring, Anomaly Detection and Forecasting

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Abstract— FinGear AI is an explainable personal financial intelligence application designed as a Financial Digital Twin for customized financial wellness, forecasting, and decision support. Users can securely save financial profiles, record income and expenses, attach PDF bills, track transactions, and recover recently deleted records. Users are classified into income tiers and transactions into Needs, Wants, and Savings. Instead of a fixed 50:30:20 rule, FinGear AI applies adaptive budgeting targets based on income level. It assesses spending distribution, cash flow, emergency reserves, debt load, saving behavior, investment consistency, and payment discipline to produce an interpretable financial health score with component-level reasons, risk alerts, and actionable recommendations. Abnormal expenditure transactions are detected, and users can confirm normal records or exclude irregular ones from the model's learning. Personalized expense and cash-flow forecasts begin with rule-based estimates for new users and transition to user-specific machine-learning and ARIMA models as transaction history grows. The Financial Digital Twin unifies profile data, transaction behavior, health indicators, anomaly signals, and forecasts to support budgeting, goal setting, AI Copilot guidance, what-if simulations, and investment decision support. Built with React, FastAPI, PostgreSQL, and machine-learning services, it offers a secure, transparent, and scalable platform for better personal financial decision making.

Keywords—Financial digital twin, explainable AI, financial health score, anomaly detection, ARIMA, random forest, personal finance management, income-tier budgeting.

I. INTRODUCTION

Financial wellness depends on a person's ability to meet current commitments, absorb financial shocks, manage borrowing and make progress toward future goals. These outcomes are not visible from income alone; a high-income individual may face unstable cash flow or inadequate reserves, while a lower-income individual may be resilient through disciplined behaviour. A useful personal-finance system must therefore analyse behaviour in context rather than impose a single generic rule.

Existing personal-finance applications largely record income, expenses and balances, offering fragmented and reactive insight. Users may see how much they spent in a category without understanding whether the pattern is sustainable, or how it affects their emergency fund, debt position or long-term goals. Generic rules such as the fixed 50:30:20 allocation cannot account for differences in income, dependents, debt obligations and financial priorities.

FinGear AI addresses this gap through an explainable Financial Digital Twin. This paper reports the Phase I assessment layer of the system, which is limited in scope to three tightly connected modules: (i) an adaptive, explainable Financial Health Score, (ii) category-specific transaction anomaly detection, and (iii) progressive expense and cash-flow forecasting. Each output is derived from data the user has supplied, is traceable to an explicit rule or model, and is designed to support the later Digital Twin, planning and advisory features of the system.

II. LITERATURE REVIEW

Uyanahewa et al. proposed WONGA, a smart personal-finance application that automates expense tracking through bill-image processing, bank-SMS extraction, natural-language processing and calendar-event expense prediction. The system used ANN models for SMS classification, OCR and NER for bill extraction, Multinomial Naive Bayes for expense categorisation, ARIMA for short-term expense forecasting, and DistilBERT with regression models for event-based expense estimation, reporting 97% SMS classification accuracy, 88% transaction-extraction accuracy and 98.93% expense-classification accuracy. Its limitations include dependence on region-specific datasets and the absence of an explainable health score, adaptive income-tier budgeting, anomaly feedback or Digital Twin modelling [1].

Hyndman and Khandakar presented automatic forecasting methods through the forecast package for R, in which exponential-smoothing or ARIMA models are automatically selected, parameterised and used to generate seasonal and non-seasonal forecasts without requiring a forecasting expert for every series. This establishes the statistical foundation used by FinGear AI's baseline-to-ARIMA transition, although the work is a general forecasting framework and does not address personal-finance context, transaction classification or explainable recommendations [2].

Anshari, Almunawar and Masri explored the use of Digital Twin concepts in robo-advisory systems, proposing that a financial Digital Twin can integrate personal data, real-time information and advanced analytics to provide dynamic, customised advice instead of one-time output. The resulting conceptual model shows that Digital-Twin-enabled robo-advisors can improve personalisation and engagement, but the study is exploratory and does not implement or evaluate a complete transaction-level, health-scoring or forecasting system [3].

Bonelli examined Digital Twin personalisation and conversational AI in robo-advisory through a scenario-based randomised 2×2 experiment involving 336 respondents with investment experience, representing Digital Twins as dynamic investor profiles incorporating goals, risk tolerance and cash-flow patterns. Results showed that Digital Twin personalisation increased perceived personalisation but also increased privacy concern, while conversational AI improved perceived interaction quality. The study measured scenario-based perceptions rather than real adoption or financial outcomes [4].

Acharya and Bhojak proposed the Financial Doppelgangers framework, integrating Digital Twins, Graph Neural Networks, Distributed Ledger Technology and Explainable AI to detect systemic risk in FinTech ecosystems, evaluated on more than 12,000 peer-to-peer lending transactions and reporting 88% early-warning accuracy, a 30% reduction in contagion lag and an AUC of 0.91. The framework targets ecosystem-level resilience rather than individual personal-finance management, and its infrastructure is more complex than required for a student-oriented application [5].

Jiang et al. proposed an investment-advisory system based on LSTM networks and multi-objective optimisation that uses historical market information, investor behaviour, financial goals and risk appetite to generate personalised portfolio strategies. Simulated results suggested more stable responses to market volatility than conventional approaches, but the work relies mainly on simulated scenarios and does not establish real-world suitability, regulatory compliance or integration with everyday spending behaviour [6].

Kong et al. reviewed Large Language Model applications in finance and investment management, categorising work involving financial text analysis, sentiment analysis, time-series forecasting, financial

reasoning and agent-based modelling. The paper concludes that LLMs can support personalised financial advice but also highlights risks of hallucination, unreliable reasoning and data-privacy concerns, motivating an approach in which generated explanations are grounded in structured profile and transaction data rather than treated as an independent source of financial truth [7].

Das, Markowitz, Scheid and Statman developed a mental-account portfolio framework combining Mean-Variance Theory with Behavioural Portfolio Theory, separating investments into goal-specific sub-portfolios and defining risk as the probability of failing to reach each goal's threshold. The study demonstrates a mathematical relationship between mental accounts, mean-variance optimisation and value-at-risk-based management, but does not incorporate real-time income, expense transactions or AI-based behaviour analysis [8].

Leukam, Koffi and Djagba proposed a goal-based portfolio-optimisation method using G-Learning and GIRL, an inverse-reinforcement-learning extension that dynamically adjusts portfolio positions to maximise value by a target date while minimising periodic contribution requirements, reporting an improvement in simulated Sharpe Ratio from 0.42 to 0.483. The results are based on simulation and do not account fully for real transaction costs, market impact or consumer deployment [9].

Feng, Li and Liu proposed a responsible-AI framework for robo-advisory and financial planning based on five principles — fiduciary duty, adaptive personalisation, technical robustness, ethics and fairness, and auditability — and presented a five-level roadmap for AI financial intermediaries, arguing that AI can improve financial access only when design safeguards prevent information asymmetry and opaque decision-making. The contribution is identified as being conceptual and policy-oriented, rather than as a fully implemented system [10].

Černevičienė and Kabašinskas conducted a systematic literature review of 138 studies on Explainable AI in finance, identifying credit management, stock prediction and fraud detection as common applications and finding that feature importance, SHAP, rule-based methods and combined explanation frameworks are frequently used to improve transparency and trust. The review synthesises prior work rather than deploying a new personal-finance system, and identifies unresolved challenges in measuring explanation quality and fairness [11].

Collectively, the surveyed literature establishes strong but separate foundations for FinGear AI. WONGA provides the closest base for transaction extraction, automated classification and ARIMA-based prediction; Hyndman and Khandakar provide the statistical foundation for automatic forecasting; the Digital Twin studies show that dynamic user representations can improve personalisation while raising privacy and trust considerations; the investment and portfolio papers contribute goal-oriented allocation and reinforcement-learning approaches, generally assuming a stable, already-funded profile; and the Explainable-AI review supports the

use of interpretable rules for the core health score alongside machine learning for anomaly detection and forecasting. No reviewed system fully combines adaptive income-tier budgeting, a connected transaction ledger, an explainable health score, per-user category-specific anomaly feedback and progressive forecasting within a single architecture — the gap that Phase I of FinGear AI addresses.

III. METHODOLOGY

The Phase I architecture consists of a React frontend and a FastAPI service layer backed by PostgreSQL, organised into two connected sections: a Financial Analytics System and a Personalized Decision Support layer. Financial profile and transaction data submitted through the frontend pass through JWT-authenticated Application Programming Interface (API) routes, after which a data-processing pipeline performs input validation, transaction categorisation, feature preparation and organisation of user history. PostgreSQL serves as the persistent, user-scoped store for profiles, transactions, health-score history, anomaly feedback and forecast records, ensuring that every analytical engine works from a single, consistent ledger rather than independently maintained copies of the same data.

Three analytical engines operate on this shared financial context. The anomaly-detection engine evaluates new spending against the user's established category-specific behaviour; the forecasting engine combines statistical baselines, Autoregressive Integrated Moving Average (ARIMA) and a user-specific Random Forest regressor according to the depth of available history; and the financial-health engine applies explainable, tier-aware weighted rules derived from the classic 50:30:20 principle, adapted per income tier. Health scores, forecasts and anomaly alerts produced by these engines are returned through the API and consumed by the Personalized Decision Support layer, which assembles them into transaction and expense-tracking views, health and forecast dashboards, and — in later phases — goal planning, what-if simulation, AI-Copilot guidance and investment decision support. The architecture therefore treats the Financial Digital Twin as a logical view over this shared analytical state rather than a separately maintained database, as illustrated in Fig. 1.

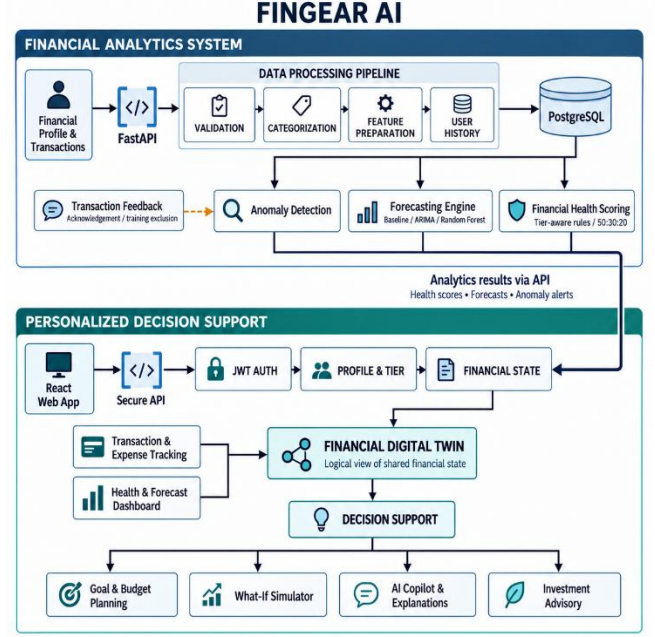


Fig. 1. FinGear AI system architecture (Financial Analytics System and Personalized Decision Support)

A. Financial Health Score

Users are classified into five income tiers — Survival, Baseline, Accumulation, Reverse Budget and Wealth Building — each with a distinct Needs–Wants–Savings (N:W:S) target ranging from 65:15:20 for Survival to 35:15:50 for Wealth Building. Let I denote average monthly income, NE needs expenditure, WA wants expenditure, SV savings, EMI mandatory debt repayment, OD outstanding debt and EF liquid emergency funds, SS Spending Sustainability, ER Emergency Resilience, DM Debt Manageability, FR Future Readiness, PD Payment Discipline Protection. The actual allocation ratios are then computed as in (1):

$$N = NE / I, \quad W = WA / I, \quad S = SV / I \quad (1)$$

Spending Sustainability (weight 30%) combines allocation fit and surplus fit, computed using (2) and (3):

$$AllocationFit = 0.25 \cdot NeedsFit + 0.30 \cdot WantsFit + 0.45 \cdot SavingsFit \quad (2)$$

$$SS = 0.75 \cdot AllocationFit + 0.25 \cdot SurplusFit \quad (3)$$

Emergency Resilience (weight 25%) compares liquid funds against a risk-adjusted target based on dependents and employment type, as defined in (4):

$$ER = \min(100, 100 \cdot EF / EmergencyTarget) \quad (4)$$

Debt Manageability (weight 20%) penalises high EMI-to-income and debt-to-annual-income ratios, as given in (5):

$$DM = 0.60 \cdot EMIScore + 0.30 \cdot DebtScore + 0.10 \cdot (OnTimeRate \cdot 100) \quad (5)$$

Future Readiness (weight 15%) evaluates saving-target achievement, goal feasibility and investment consistency, as given in (6):

$$FR = 0.50 \cdot \text{SavingTargetAchievement} + 0.30 \cdot \text{GoalFeasibility} + 0.20 \cdot \text{InvestmentConsistency} \quad (6)$$

Payment Discipline and Protection (weight 10%) evaluates on-time payments, planning status and insurance coverage, as given in (7):

$$PD = 0.60 \cdot \text{PaymentScore} + 0.25 \cdot \text{PlanningScore} + 0.15 \cdot \text{ProtectionScore} \quad (7)$$

The overall raw score combines all five weighted components as shown in (8):

$$\text{RawScore} = 0.30 \cdot SS + 0.25 \cdot ER + 0.20 \cdot DM + 0.15 \cdot FR + 0.10 \cdot PD \quad (8)$$

Two safety guardrails prevent strong performance in one area from concealing critical risk: the final score cannot exceed 59 when monthly surplus is negative, and cannot exceed 49 when EMI reaches 40% of income. Scores are graded Financially Healthy (80–100), Stable/Improving (60–79), Needs Attention (40–59) or Financially Vulnerable (0–39). The engine additionally raises risk flags whenever emergency reserves cover less than one month of essential outflow, a bill or EMI has been missed within the last 90 days, EMI pressure reaches at least 35% of income, or outstanding debt exceeds 1.2 times annual income, giving the user explicit, actionable warnings rather than a single opaque number.

A separate Data Confidence measure, given in (9), is computed independently of the health score itself, so that incomplete records reduce certainty rather than silently lowering the user's result:

$$\text{DataConfidence} = 0.40 \cdot \text{HistoryCompleteness} + 0.40 \cdot \text{CategorisationCompleteness} + 0.20 \cdot \text{IncomeVerification} \quad (9)$$

The system reports High Confidence above 80, Moderate Confidence between 50 and 79, and a Provisional Score below 50 or when history is insufficient. Finally, an adaptive recommendation engine converts the score into practical next steps: excess discretionary spending is redirected gradually rather than all at once, so that the recommended monthly reduction is capped at one third of the excess or 5% of income, whichever is smaller, and any emergency-fund shortfall or required increase in savings is expressed as a concrete monthly target rather than a generic instruction.

B. Transaction Anomaly Detection

Each user maintains an independent anomaly-detection model for every spending category — Food, Transport, Housing, Utilities, Shopping, Entertainment and others — so the same transaction amount may be considered normal in one category and unusual in another for the same person. When a user has insufficient history in a category, a cold-start process based on robust baseline rules and category-level median behaviour is used, allowing anomaly

evaluation to begin even with limited data. As more transactions accumulate, the model transitions to a machine-learning ensemble combining category-level clustering (K-Means) with autoencoder-based pattern reconstruction, so that both distance-based and reconstruction-error-based signals contribute to the final anomaly score.

For each evaluated transaction, the module returns the spending category, a normalised anomaly score, a severity level, the current model's maturity status and a short explanation of which behavioural signal triggered the flag. Rather than automatically rejecting a transaction, the module functions as a decision-support layer: the user can confirm that a flagged transaction is normal, in which case it is folded back into the category's learned baseline, or exclude it from future model training, in which case the record remains visible in the ledger for bookkeeping but no longer influences what the model considers typical for that category. This design allows a single unusual but legitimate event — such as a medical bill or emergency repair — to be handled without permanently distorting the model's understanding of the user's everyday spending, and because models are isolated per user and per category, one person's behaviour never affects another user's anomaly baseline.

C. Progressive Expense Forecasting

The forecasting engine selects a model according to the amount and quality of available personal history, moving through three progressive stages rather than committing to a single model for every user. A statistical/rule-based baseline is used for newly registered users, for categories with fewer than seven distinct expense days, or whenever recent history is incomplete; it produces a cautious estimate from recent average expenses and observed transaction frequency, together with an explicit maturity label so the user understands how much to trust it. Once at least seven distinct expense days are available, a user-specific Random Forest regressor is trained on the individual's own transaction history to capture nonlinear relationships that a simple average cannot represent — for example, higher spending on particular days of the month, weekend spending spikes, category-specific patterns, and interactions between time period and category.

As multi-month time-series data accumulates, a per-user ARIMA model is additionally trained to capture trend and temporal dependency between successive periods, complementing the Random Forest model's ability to capture nonlinear, multi-feature relationships. Before any model is trained, raw transactions are preprocessed into date, type, category, amount, day/week/month, recurrence and recent-trend features drawn from the same validated ledger used by the health-score and anomaly-detection engines, so that all three modules remain consistent with one another. Daily forecasts support short-term planning by estimating the expected expense for upcoming days, while monthly and category-level forecasts aggregate these estimates together with recurring transactions to project cash-flow movement, as shown in (10) and (11):

$$ProjSurplus = ProjIncome - ProjNeeds - ProjWants - ProjDebtPayments \quad (10)$$

$$ProjSavingsCapacity = \max(0, ProjSurplus) \quad (11)$$

Every forecast is labelled provisional, developing or mature according to the depth of the user's transaction history, the number of distinct expense days, category completeness and the stability of observed spending — so that a prediction built from a few days of data is never presented with the same certainty as one built from an established, multi-month time series. Forecasts are recalculated automatically whenever a new transaction, a recurring-transaction change, an income adjustment, a restored record or a model-training exclusion alters the underlying ledger, so the projected surplus and category-level estimates always reflect the user's current financial condition rather than a stale snapshot.

IV. RESULT AND ANALYSIS

The three Phase I modules were evaluated independently using module-specific benchmarks.

A. Financial Health Score

The scoring engine was tested across all five income tiers and achieved 100% alignment with the defined tier-based rules. Table I summarises representative results. The critical-debt guardrail also achieved a 100% trigger rate: whenever the tested debt-to-income ratio exceeded 40%, the score was correctly capped at 49 or below.

TABLE I. HEALTH SCORE VALIDATION ACROSS INCOME TIERS

Income Tier	Monthly Income	Score	Assessment
Survival	₹22,000	38/100	Vulnerable
Baseline	₹42,000	72/100	Improving
Accumulation	₹72,000	92/100	Healthy
Reverse Budget	₹1,15,000	97/100	Healthy
Wealth Building	₹2,40,000	97/100	Healthy

B. Anomaly Detection

The ensemble was evaluated on 300 labelled transactions (240 normal, 60 anomalous), achieving 95.67% classification accuracy. All 240 normal transactions were correctly classified and 47 of the 60 anomalies were detected, giving 100% precision, 78.33% recall, an F1-score of 0.8785 and a Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) of 0.9992. No false positives occurred, meaning every flagged transaction in the evaluation was genuinely anomalous, though 13 anomalies remained undetected — indicating a trade-off between alert reliability and recall, as shown by the score distribution and confusion matrix in Fig. 2 and the flagged transactions in Fig. 3.

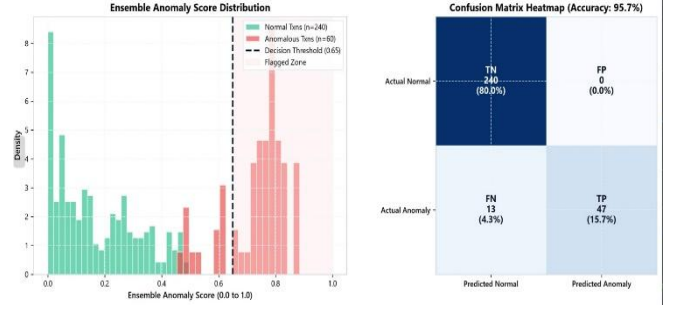


Fig. 2. Anomaly score distribution and classification confusion matrix

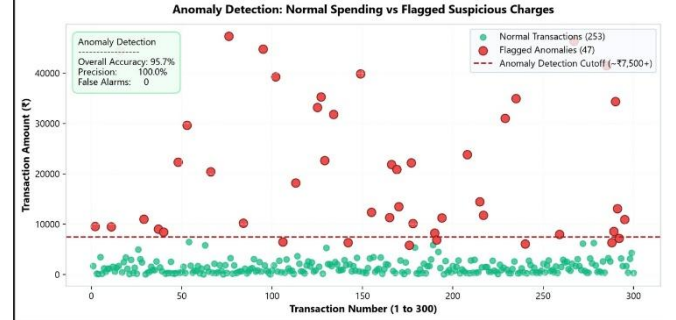


Fig. 3. Transaction spending distribution with flagged anomalies

C. Expense Forecasting

The Random Forest expense forecaster was evaluated using an 83-day rolling walk-forward back-test, achieving 82.94% forecasting accuracy, a mean absolute error of ₹210.39 per day, a root-mean-squared error of ₹295.88 and an R^2 of 0.5624. Directional accuracy (predicting whether spending would rise or fall) was lower at 48.78%, showing that the model estimates spending magnitude more reliably than short-term direction. Average response time was 44.86 ms under validation conditions, and Fig. 4 compares actual against AI-predicted daily expenses over the evaluated window.

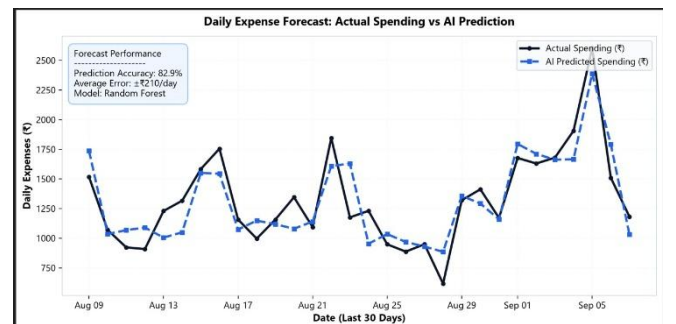


Fig. 4. Actual versus AI-predicted daily expenses over a 30-day window

Overall, the three modules — 100% rule alignment, 95.7% anomaly-detection accuracy and 82.9% forecast accuracy — represent different performance measures and should not be combined into a single score. Financial-health scoring demonstrated consistent application of the specified tier rules and debt safeguards across every tested scenario. Anomaly detection achieved high classification accuracy and produced no false alarms, although its 78.33% recall shows that a portion of unusual transactions

were missed, highlighting a trade-off between alert reliability and detection coverage that future retraining and threshold tuning could improve. Expense forecasting exceeded its stated benchmarks for accuracy, average error and R^2 , but its comparatively low directional accuracy indicates that estimating the magnitude of future expenditure and anticipating its day-to-day movement remain distinct capabilities; the forecasting output is therefore best interpreted alongside its error measures rather than through the headline accuracy percentage alone. Together, the results demonstrate that rule-grounded explainability and adaptive machine learning can coexist within a single personal-finance system, and that all evaluated benchmarks for the three Phase I modules were met within the tested conditions. Broader validation across additional users, spending categories and longer time horizons would help establish how consistently these results extend beyond the evaluated cases.

V. CONCLUSION

This paper presented the Phase I financial-assessment layer of FinGear AI, comprising an adaptive, explainable Financial Health Score, a per-user category-specific anomaly-detection engine, and a progressive baseline-to-ARIMA-to-Random-Forest forecasting pipeline. Income-tier classification ensures that users with different financial circumstances are evaluated against suitable targets rather than a single generic rule, while explicit component-level formulae keep every score, alert and forecast traceable to interpretable evidence. Validation results — 100% tier-rule alignment, 95.67% anomaly-detection accuracy with 100% precision, and 82.94% forecasting accuracy — support the feasibility of this approach. This foundation is designed to support the later Digital Twin features of FinGear AI, including goal and budget planning, what-if simulation, AI-Copilot guidance and investment decision support.

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